Nova School of Business and Economics

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Case on "Bond Portfolio Management"

Course: Fixed Income
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Date: 13.03.2026

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Assumptions

The following parameters are used consistently throughout all exercises and serve as the basis for all calculations presented in this report.

Table 1: Interest Rate Assumptions

Parameter	**Symbol**	**Value**	**Notes**
Initial flat rate (continuous)	r_0	2.00%	At $t = 0$; Exercise 1
Shifted flat rate (continuous)	r_1	2.40%	After rate shift; Exercises 1-Q2 and 2

Table 2: Liability Assumptions (Guaranteed Investment Contracts)

Parameter	**Symbol**	**Value**	**Unit**	**Notes**
GIC 1 payment amount	L_1	100,000	€	Paid at $T = 2$
GIC 2 payment amount	L_2	110,000	€	Paid at $T = 3$
GIC 1 maturity	T_1	2	years	
GIC 2 maturity	T_2	3	years	

Table 3: Bond Coupon Assumptions

Parameter	**Symbol**	**Value**	**Notes**
CBB 5% annual coupon rate	c	5.00%	5-year coupon-bearing bond; annual coupon; Exercise 1
CBB 2% annual coupon rate	c_2	2.00%	2-year coupon-bearing bond; annual coupon; Exercise 2

Table 4: Portfolio Assumption

Parameter	**Symbol**	**Value (€)**	**Notes**
Portfolio value at $t = 1$	$V_A(1)$	202,478.70	Given; used in Exercise 2

Exercise 1

Question 1: Immunizing Portfolio at $t = 0$

Table 5: Present Value of Liabilities at $t = 0$ ($r_0 = 2.00\%$ continuous)

Liability	**Amount (€)**	**Maturity T**	**Discount Factor**	**PV(L) (€)**
GIC 1	100,000.00	2	0.9608	96,078.94
GIC 2	110,000.00	3	0.9418	103,594.10
Total $V_L(0)$				**199,673.04**

Table 6: Macaulay Duration of the Liability Portfolio

Liability	T	**PV(L) (€)**	$T \times$ **PV(L)**
GIC 1	2	96,078.94	192,157.89
GIC 2	3	103,594.10	310,782.30
Total		199,673.04	502,940.18
Duration (Liabilities)			**2.5188 years**

Table 7: CBB 5% — Cash Flows, Price, Duration and Convexity at $t = 0$

Year	**CF (per face)**	**Disc. Factor**	**PV(CF)**	$t \times$ **PV(CF)**	$t^2 \times$ **PV(CF)**
1	5.00%	0.9802	0.0490	0.0490	0.0490
2	5.00%	0.9608	0.0480	0.0961	0.1922
3	5.00%	0.9418	0.0471	0.1413	0.4238
4	5.00%	0.9231	0.0462	0.1846	0.7385
5	105.00%	0.9048	0.9501	4.7504	23.7520
Price (CBB 5%)			**1.1404**		
Duration (CBB 5%)				**4.5787 years**	
Convexity (CBB 5%)					**22.0590**

Table 8: Asset Properties and Duration-Matching Weights at $t = 0$

Asset	**Maturity (yrs)**	**Price (per face)**	**Duration**	**Convexity**
FRB	4	1.0000	1.0000	1.0000
CBB 5%	5	1.1404	4.5787	22.0590
Duration (Liabilities)				**2.5188 years**
Weight in FRB				**57.56%**
Weight in CBB 5%				**42.44%**

Table 9: Immunizing Portfolio Composition at $t = 0$

Asset	**Market Value (€)**	**Price (per face)**	**Face Value** N **(€)**
FRB	114,929.71	1.0000	114,929.71
CBB 5%	84,743.33	1.1404	74,311.96
Total $V_L(0)$	**199,673.04**		

At time $t = 0$, the term structure of interest rates is flat at 2.00% with continuous compounding. The insurance company has issued two Guaranteed Investment Contracts. GIC1 requires a payment of €100,000 in 2 years and GIC2 requires a payment of €110,000 in 3 years. The present values are €96,078.94 for GIC1 and €103,594.10 for GIC2, for a total liability value of €199,673.04. The liability portfolio has a Macaulay duration of 2.5188 years.

The company immunizes these liabilities with two assets. The first is a 4-year Floating Rate Bond (FRB) that resets its coupon each year, so its price stays at par and its duration equals 1 year. The second is a 5-year Coupon Bearing Bond (CBB) with a 5% annual coupon. The CBB trades at €1.1404 per unit of face value and has a Macaulay duration of 4.5787 years. The analysis ignores convexity.

To match the liability duration, the company applies two conditions. It invests the full present value of the liabilities and matches the duration of the asset portfolio to 2.5188 years. Solving the resulting two-equation linear system gives a portfolio weight of 57.56% in the FRB and 42.44% in the CBB, based on market values. These weights lead to market value allocations of €114,929.71 in the FRB and €84,743.33 in the CBB. Because the FRB trades at par, its face value equals €114,929.71. The CBB face value equals €74,311.96, obtained by dividing €84,743.33 by the price of €1.1404 per unit. The two allocations add to €199,673.04, so the asset portfolio matches the present value of the liabilities.

Question 2: Terminal Wealth at $T = 3$ (Rate Shift to $r_1 = 2.40\%$)

Table 10: FRB — Cash Flows and Future Value at $T = 3$

CF	Paid at t	Amount (€)	Reinv. Years	FV Factor	FV at T=3 (€)
$c(1)$	1	2,321.73	2	1.0492	2,435.90
$c(2)$	2	2,791.68	1	1.0243	2,859.49
$c(3)$	3	2,791.68	0	1.0000	2,791.68
Principal	3	114,929.71	0	1.0000	114,929.71
FV(FRB) at $T = 3$					**123,016.78**

Table 11: CBB 5% — Cash Flows and Future Value at $T = 3$

CF	Paid at t	Amount (€)	Reinv. Years	FV Factor	FV at T=3 (€)
$c(1)$	1	3,715.60	2	1.0492	3,898.30
$c(2)$	2	3,715.60	1	1.0243	3,805.85
$c(3)$	3	3,715.60	0	1.0000	3,715.60
MV at T=3	3	77,998.19	0	1.0000	77,998.19
FV(CBB 5%) at $T = 3$					**89,417.93**

Table 12: Terminal Wealth at $T = 3$ after Parallel Rate Shift

Item	**Value (€)**
FV(FRB) at $T = 3$	123,016.78
FV(CBB 5%) at $T = 3$	89,417.93
Total Asset FV	**212,434.71**
Less: GIC 1 reinvested to $T = 3$	−102,429.03
Less: GIC 2 paid at $T = 3$	−110,000.00
Terminal Wealth (Assets − Liabilities)	**5.67**

To test how robust this immunization is, assume interest rates jump at once to 2.40% with continuous compounding immediately after the portfolio starts and then stay at 2.40%. The first coupon on the FRB at $t = 1$ was set at issue using the original 2.00% rate. This gives $c(1) = e^{0.02} - 1 \approx 2.0201\%$ of face value. From $t = 1$ onward coupons reset to the new rate, so $c(2)$ and $c(3)$ both equal $e^{0.024} - 1 \approx 2.4290\%$ of face value. All cash flows the FRB pays before $t = 3$ are reinvested at 2.40%. At $t = 3$ the FRB redeems at par. The total future value then equals €1.0704 per unit of face value, which for the full holding equals €123,016.78.

For the CBB, the investor receives three annual coupons of 5% on a face value of €74,311.96. Each coupon equals €3,715.60 and is reinvested at 2.40% until $t = 3$. The remaining two cash flows, at $t = 4$ and $t = 5$, are sold at $t = 3$ at their market value. Discounting them at the new 2.40% rate gives a value of €77,998.19 at $t = 3$. The total value of the CBB position at $t = 3$ equals €89,417.93.

The total asset value at $t = 3$ equals €212,434.71, as the sum of the FRB and CBB positions. The liabilities at $t = 3$ are GIC1 and GIC2. GIC1 pays €100,000 at $t = 2$, then grows for one year at 2.40% to €102,429.03 at $t = 3$. GIC2 pays €110,000 at $t = 3$. Together these liabilities equal €212,429.03. Terminal wealth, defined as assets minus liabilities, equals €5.67. This value lies close to zero and shows that the immunization works. The small positive amount arises from the higher convexity of the asset portfolio compared with the convexity of the liability stream, so the portfolio gains slightly when interest rates shift in parallel.

Exercise 2: Portfolio Rebalancing at $t = 1$

Table 13: Remaining Liabilities at $t = 1$ ($r_1 = 2.40\%$ continuous)

Liability	**Remaining** T	**PV(L) (€)**	$T \times$ **PV(L)**	$T^2 \times$ **PV(L)**
GIC 1 (1 year left)	1	97,628.57	97,628.57	97,628.57
GIC 2 (2 years left)	2	104,844.72	209,689.43	419,378.87
Total $V_L(1)$		**202,473.29**	307,318.00	517,007.44
Duration (Liabilities)				**1.5178 years**
Convexity (Liabilities)				**2.5535**

Table 14: CBB 5% — Cash Flows, Price, Duration and Convexity at $t = 1$

Year	CF (per face)	Disc. Factor	PV(CF)	$t \times$ PV(CF)	$t^2 \times$ PV(CF)
1	5.00%	0.9763	0.0488	0.0488	0.0488
2	5.00%	0.9531	0.0477	0.0953	0.1906
3	5.00%	0.9305	0.0465	0.1396	0.4187
4	105.00%	0.9085	0.9539	3.8155	15.2622
Price (CBB 5%)			**1.0969**		
Duration (CBB 5%)				**3.7372 years**	
Convexity (CBB 5%)					**14.5142**

Table 15: CBB 2% — Cash Flows, Price, Duration and Convexity at $t = 1$

Year	CF (per face)	Disc. Factor	PV(CF)	$t \times$ PV(CF)	$t^2 \times$ PV(CF)
1	2.00%	0.9763	0.0195	0.0195	0.0195
2	102.00%	0.9531	0.9722	1.9444	3.8888
Price (CBB 2%)			**0.9917**		
Duration (CBB 2%)				**1.9803 years**	
Convexity (CBB 2%)					**3.9409**

Table 16: Asset Properties at $t = 1$

Asset	Maturity (yrs)	Price (per face)	Duration	Convexity
FRB	3	1.0000	1.0000	1.0000
CBB 5%	4	1.0969	3.7372	14.5142
ZCB	0.5	0.9881	0.5000	0.2500
CBB 2%	2	0.9917	1.9803	3.9409

Table 17: Portfolio at $t = 1$ Before Rebalancing

Asset	Face Value N (€)	Price (per face)	Market Value (€)	Weight
FRB	114,929.71	1.0000	114,929.71	56.76%
CBB 5%	74,311.96	1.0969	81,511.65	40.26%
Cash (coupon)	—	—	6,037.34	2.98%
Total			**202,478.70**	**100.00%**
Duration (Assets): 2.0721 years	**Convexity (Assets):** 6.4106			

Table 18: Rebalanced Portfolio at $t = 1$

Asset	**Old FV N (€)**	**Weight**	**New MV (€)**	**New FV N (€)**	**Trade (€)**
FRB	114,929.71	50.00%	101,239.35	101,239.35	−13,690.36
CBB 5%	74,311.96	21.17%	42,860.67	39,074.91	−38,650.98
ZCB	0	23.25%	47,084.12	47,652.53	+47,084.12
CBB 2%	0	5.58%	11,294.56	11,388.84	+11,294.56
Cash deployed					−6,037.34
Total		**100.00%**	**202,478.70**		0.00

Table 19: Verification of Immunization Conditions at $t = 1$

Condition	**Target**	**Achieved**	**Difference**
Duration (Assets) match Duration (Liabilities)	1.5178	1.5178	0.0000
Convexity (Assets) $\geq$ Convexity (Liabilities)	2.5535	3.8503	+1.2969
Budget constraint (sum of weights = 1)	1.0000	1.0000	0.0000
Max weight $\leq$ 50%	50.00%	50.00%	0.00%

At time $t = 1$, the interest rate stays constant at 2.40% continuous compounding, and the portfolio value is €202,478.70. The insurance company will rebalance the portfolio to immunize the remaining liabilities of €100,000 due in 1 year and €110,000 due in 2 years. The rebalancing includes the two existing bonds (FRB with 3 years remaining and CBB 5% with 4 years remaining) and two new assets (a 6-month ZCB and a 2-year CBB paying 2% annually). The objective is to minimize transaction costs, approximated by minimizing the squared differences between the new and old face values of the assets in solver. Additionally, we fully reinvest the portfolio value to match the liability duration of 1.5178 years, satisfy a convexity condition of at least 2.5535 years^2, and ensure no asset exceeds 50% weight of the portfolio by market value.

The optimal new face values are FRB at €101,239.35, CBB 5% at €39,074.91, 6-month ZCB at €47,652.53, and 2-year CBB 2% at €11,388.84. These correspond to market values and weights of €101,239.35 (50.00%), €42,860.67 (21.17%), €47,084.12 (23.25%), and €11,294.56 (5.58%), adding up to the total portfolio value of €202,478.70.

This allocation achieves an exact match with the target duration of 1.5178 years, while the portfolio convexity of 3.8502 years^2 slightly exceeds the liability convexity of 2.5535 years^2 (permitted by the inequality constraint). As a result, the portfolio provides robust immunization against both small and larger interest rate shifts. Transaction costs are minimized through the optimization objective, which penalizes large deviations from the original face values of €114,929.71 for the FRB and €74,311.96 for the 5% CBB. The portfolio remains well diversified, including the FRB at the 50% cap and shorter-duration assets such as the ZCB, thereby providing balance and stability under the given constraints and ensuring the portfolio closely tracks the liabilities over the remaining horizon.

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